Ahmedabad University

Subject - Digital Image Processing

8. Vehicle Orientation Estimation Using Conventional Image Processing

Mid-Semester Report

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Repo Link: https://github.com/encrypted-official/DIP_2026_Project8_Group1

Group - 1

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1. Motivation

Drone footage is now used quite often to study traffic in Indian cities. From such footage it is useful to know not only where a vehicle is, but also which way it is pointing. This helps in studying lane flow, spotting vehicles going the wrong way and checking how vehicles are parked. Most recent work does this with deep networks that need a GPU and a large amount of labelled data.

We wanted to see how far the basic tools from this course can go on the same task, such as filtering, edge detection, PCA and the Hough transform. These methods are cheap to run and each step can be checked by hand, so when they fail it is easier to see why. The task is still hard. Vehicles in aerial images are often less than 30 pixels long, many are parked close together, and shadows, trees and road markings add edges that do not belong to the vehicle.

2. Problem Definition

Given a crop of a single vehicle from an aerial image, estimate the angle of its body axis using only conventional image processing, and compare it with the annotation in the DRASHTI-HaOBB dataset. Since our methods cannot tell the front of the vehicle from the back, the heading angle θ from the dataset is folded into an undirected axis angle φ before comparison:

$$r = \theta \bmod 180°, \qquad \varphi = \min(r,\ 180° - r), \qquad \varphi \in [0°,\ 90°]$$

For a predicted axis p the error is $e = |\varphi_p - \varphi|$ in degrees. The same rule is applied to the ground truth and to the predictions.

3. Literature Review

Over the first three weeks we read papers in two groups: work on vehicle orientation in general, and papers behind the exact steps we use in our code.

- **Zhou et al. (2017)** used Canny edges and the Radon transform to find vehicle orientation in aerial images. This is the closest work to ours and is the reason we started with edge based methods.
- **Rybski et al. (2010)** used HOG features to classify vehicle orientation into bins. We kept HOG as an option for later.
- **Hu (1962)** showed that the second order central moments of a shape give its orientation. This is the same idea as our PCA step, so we do not count PCA and moments as two separate methods.
- **Duda and Hart (1972)** gave the rho-theta form of the Hough transform, and **Matas et al. (2000)** gave the probabilistic version that OpenCV's HoughLinesP is based on.

- **Canny (1986)** and **Zuiderveld (1994)** describe the edge detector and CLAHE that form our preprocessing.
- **Mardia and Jupp (2000)** explain how to average axial data, where α and α + 180° are the same. We use their doubled angle trick when combining Hough lines.
- **Xia et al. (2018)** built the DOTA dataset, which helped us understand the oriented box format used in DRASHTI-HaOBB.

Most of the work we found either uses learning or tests on clean scenes with large vehicles. We did not find much on very small vehicles in dense Indian traffic using only classical methods, which is the gap we are looking at.

4. Method

Each vehicle is cut out using the axis aligned box around its four OBB corners with 10% padding. The crop is not rotated, so the angle we want to find stays in the image. The estimators only see the crop pixels and never the annotation angle. Preprocessing is the same for both methods: grayscale, 3×3 Gaussian blur, CLAHE, and Canny with thresholds 50 and 150.

Method A, PCA on edge pixels. The edge pixel coordinates p_i are treated as a point cloud and their covariance matrix is computed:

$$C = (1/N)\ \Sigma\ (p_i - \mu)(p_i - \mu)^T$$

The eigenvector of the largest eigenvalue λ_1 gives the axis. If $N < 20$ or $\lambda_1/\lambda_2 < 1.05$ the cloud is too small or too round, and the method returns no prediction instead of a random angle.

Method B, Hough lines. HoughLinesP is run on the same edge map and short segments are removed. Each remaining segment has angle α_i and length w_i. Because an axis has no direction, the angles are doubled before averaging and then halved:

$$\varphi_p = \tfrac{1}{2} \cdot atan2(\ \Sigma\ w_i\ sin\ 2\alpha_i\ ,\ \ \Sigma\ w_i\ cos\ 2\alpha_i\ )$$

For evaluation we report coverage (share of crops that got a prediction), mean absolute error (MAE), median error, 90th percentile error, and the share of crops within 10°. Missing predictions are counted in coverage and not quietly dropped.

5. Results

We ran both methods on the first 40 images of the validation split. This gave 1822 vehicle crops after removing difficulty 3 objects (vehicles cut at the image edge). These are early numbers on a small sample, not a final benchmark.

Table 1: Overall results on the validation sample

Method	Valid / Total	Coverage (%)	MAE (°)	Median (°)	P90 (°)	Within 10° (%)
PCA on edges	1820/ 1822	99.89	11.15	6.282	27.212	65.22
Hough lines	1819/ 1822	99.84	14.103	6.819	39.133	62.67

- PCAhas the lower error, mostly the small ones where no long straight edge is found.
- Both methods get clearly better as the crop gets bigger. Below 20 px the MAE of both is above 10°, which is close to guessing.
- Looking at the overlays, the common failures are shadows falling along one side of the car, road markings and kerbs inside the crop, and a nearby parked vehicle whose edges get mixed in.
- The P90 error is above 30° for both, so a small group of crops is predicted almost at right angles to the truth. In many of these the edges of the roof and windscreen are stronger than the edges along the body.



Figure 1: Failure cases for PCA and Hough

6. Next Plan

- Add a third method that builds a foreground mask with Otsu thresholding and then uses the minimum area rectangle and image moments, where $\tan 2\varphi = 2\mu_{11} / (\mu_{20} - \mu_{02})$.
- Run a small ablation on Canny thresholds, CLAHE on or off, and blur size to see how much each step matters.
- Try to tell front from back using intensity differences along the estimated axis, so we can move from the axis to the full 0° to 360° heading.

- Run on the full validation split and use the train_original annotations for tuning the parameters.
- Do a proper error analysis by crop size, shadow and occlusion, and add class wise results where the sample is big enough.

7. Individual Contribution

- **Vehan Gajjar:** wrote the label parser and crop extraction, handled the metadata file and wrote the dataset notes.
- **Vedant Solanki:** wrote the shared preprocessing (blur, CLAHE, Canny) and the PCA estimator with its rejection rules.
- **Abhi Jain:** wrote the Hough line estimator with the doubled angle averaging and the method comparison notes.
- **Krish Prajapati:** wrote the angle folding helpers and their unit tests, the evaluation script and the overlay plots.
- **All members:** literature review in Weeks 1 to 3, weekly reports and the mid-semester slides.

References

[1] Zhou et al., 2017. [Add the full reference from the Week 1 notes]

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[8] K. V. Mardia and P. E. Jupp, Directional Statistics, John Wiley and Sons, 2000.

[9] G.-S. Xia et al., "DOTA: A Large-Scale Dataset for Object Detection in Aerial Images," in Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3974-3983, 2018.

[10] DRASHTI-HaOBB dataset, Zenodo, https://zenodo.org/records/18278989